Item 11: learned necessary-condition prefilter on x_R for subspace bases

Assessed 2026-10-18. Measured (toy scale, host L0). Protocol PROTOCOL.md frozen before runs. Cards CARD-BLINDSPOT-11-01, −02, −04, −06, −07 receipted; −03 and −05 incomplete.

Verdict

INCONCLUSIVE (incomplete run: 13 of 16 cell-populations), every measured cell on the MATCHES side. Level: primitive. Not an improvement. The frozen rule applied to the 13 cells on disk gives MATCHES in both populations: no learned filter (logistic regression, depth-3 boosted trees) beats the trace filter by 5 points at zero false refutations with the permutation null excluded. Missing: (n=19, m=3, S), (n=23, m=2, U), (n=29, m=3, S), because cards 03 and 05 returned before finishing. U cannot reach ADVANCES with one missing cell; S could only if both missing S cells passed. Gap to matched rho: unchanged (this stage diagnostic computes no S, ratio or GAE).

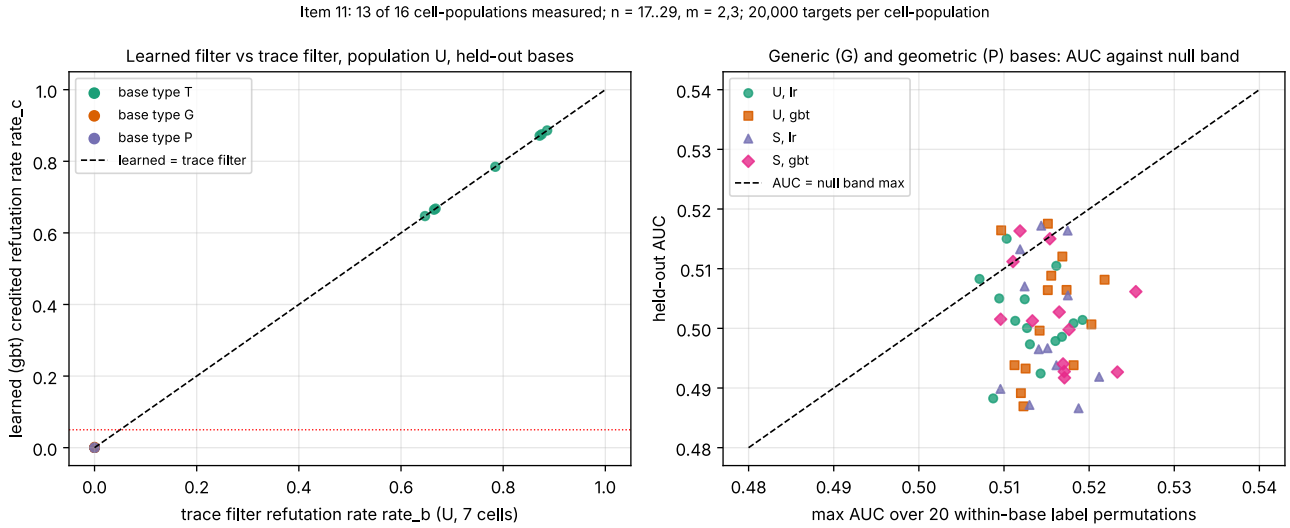


Figure 1: Left: credited refutation rate of the learned filter against the trace filter (U, gbt, 7 cells, held-out bases); trace-zero bases lie on the diagonal (the learner copies arm b), generic and geometric bases sit at 0. Right: held-out AUC against the maximum of 20 within-base permutation AUCs on G and P bases (all points near or below the diagonal). Source plot_prefilter.py, data raw/summary_rows.tsv.

Numbers (measured; 104 summary rows)

pop, type	max delta	max rate_c	max AUC - null max	false refutations
U, T	0.0000	0.886	+0.433	0
U, G	+0.0021 [0.0009, 0.0035]	0.0021	+0.0068	34
U, P	+0.0004 [0, 0.0013]	0.0004	+0.0024	41
U, ALL	−0.265 to −0.465	0	+0.208	75
S, T/G/P/ALL	0.0000	0	<= +0.0044	1/43/30/74

Threshold for a pass: delta >= 0.05 with lower interval end > 0, roc0 and AUC above the null band, planted_trace control passed. The best row (delta 0.0021) is about 24 times below the threshold.

Controls

planted_trace: AUC 1.0, calibrated refutation 1.0, 13 of 13 cells, both models. planted_selector: gbt AUC 1.0 everywhere. Permutation null: G/P AUC within 0.007 of null max. Arm (b) false refutations: 0. Selftest: 0 failures, decomposer equals brute force on every point of E(F_2^11), E(F_2^13).

Cost (section 5a pricing at 0, 0.1, 1 per unpriced word operation)

Net work of adding the gbt filter is negative at all three weights in every cell: −0.0010 (n=29, m=2, U, w=0) to −2.31 of decomposer work (n=17, m=3, S, w=1).

Deviations

Cards 03 and 05: no receipt, binaries left in target/blindspots-11, cards now fail lint, check-relay could not run. Raw files n19-m3-U.jsonl and n29-m3-U.jsonl are complete (done lines) but unreceipted. Card 07 ran before 03 and 05 finished (13

input files); receipts are write-once. Relays for 01, 02, 04, 06, 07 verified by sha256 against receipts and raw files, not re-saved as full text.

Caveats

n = 17 to 29 only; two model classes and the listed feature families only; no wall-time claim; U alone re-derives the trace, S is the deployable population and shows no learnable signal. See README.md.